

Function pointers

Function pointers are pointers, i.e. variables, which point to the address of a function.

```
// Define a function pointer
int (*pt2Function) (float, char, char);
```

Note: Thus functions and function pointers with different calling convention (argument order, arguments type or return type is different) are incompatible with each other.

Assign an address to a Function Pointer

It's quite easy to assign the address of a function to a function pointer. You simply take the name of a suitable and known function or member function.

It's optional to use the address operator & in front of the function's name.

```
//Assign an address to the function pointer
intDoIt (float a, char b, char c){ return a+b+c; }
pt2Function = &DoIt; // assignment
```

Example:

```
intaddC(charx,char y){
returnx+y;
}
intsubC(charx,char y){
return x-y;
}
intmulC(charx,char y){
return x*y;
}
intdivC(charx,char y){
return x/y;
}
intmodC(charx,char y){
returnx%y;
}
```

```
//array of pointer to functions that receive two chars
and returns int
```

```
int (*arrpf[])(char,char) = { addC ,subC,mulC,divC,modC};
```